

Vignesh Karuppasamy D M

Design Verification Engineer

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SUMMARY

Highly driven and detail-oriented Electrical and Computer Engineer with a strong foundation in digital design, semiconductor technologies, and FPGA/ASIC development. Completed a Master of Engineering at the University of Waterloo, focusing on Verilog/SystemVerilog-based RTL design, hardware verification (UVM/SystemVerilog), and the integration of ML into EDA flows to improve chip design outcomes.

TECHNICAL SKILLS

Programming Languages: Proficient with C, Python, SystemVerilog, Verilog, UVM, MATLAB. Well-versed in Linux/Windows OS

Management/Development Tools: Microsoft Office Suite (Excel, PowerPoint, Word, Teams, Outlook), Report Writing, Agile Methodology, Scrum, Technical Documentation, JIRA, Git

Soft Skills & Interpersonal Strengths: Effective Stakeholder Management, Team Collaboration, Analytical Thinking, Time Management, Strong Communication Skills, Problem-solving Skills and Analytical Skills, Project Coordination and Management

EDA & Hardware Tools: Xilinx Vivado, Intel Quartus Prime, Cadence Virtuoso, Verilator, PYNQ FPGA

FPGA & ASIC Development: RTL Design, FPGA CAD Flow, Synthesis, Functional Verification, Pipelining, Scheduling, Timing Analysis, Power & Area Optimization

Design Verification: UVM Testbench Development (agents, drivers, monitors, sequencers), Scoreboards & Checkers, Constrained-Random Test Generation, Functional & Code Coverage (covergroups, coverpoints, crosses), Assertions (SVA), Directed vs. Random Stimulus

ML for Chip Design: Regression using Scikit-learn, Convex Optimization with CVXPY, Neural Networks in PyTorch, Bayesian Optimization with Optuna and Ax/BoTorch, Evolutionary Algorithms with CMA-ES/NSGA-II/NSGA-III in pymoo, Reinforcement Learning with REINFORCE and PPO

ACCOMPLISHMENTS

- Spot Award, *Visteon India* – Recognized for resolving a critical issue (confidential) with excellence and timely delivery
- *Smart Elector Validation System Assisting the First Polling Officer* – Recognized at by Sony India & IIT Madras and the Grand Ideas Challenge at the SERB-INAE Conclave, IIT Kanpur
- *AI-Based Interactive Smart Robot for Elderly Health Monitoring*, published in [IEEE Xplore](#) (Scopus-indexed)
- *Cadence Certified* in SystemVerilog for Design and Verification

EDUCATION

University of Waterloo, Waterloo, Canada

Jan 2025 - Dec 2025

Master of Engineering – Electrical and Computer Engineering | 84.88%

- Relevant Coursework: Register-Transfer-Level Digital Systems, Computer Architecture and Organization, Microelectronic Fabrication Technology, ML for Chip Design

St. Joseph's College of Engineering, Chennai, India

Aug 2019 - Jun 2023

Bachelor of Engineering – Electronics and Communication Engineering | 92.6%

PROJECTS

High-Performance Systolic Array Matrix Multiplier on FPGA

Tech Stack: Verilog, Vivado, BRAM/DSP Optimization, Python, PYNQ FPGA

- Designed and implemented a parametric systolic array and Processing Elements (PEs) in Verilog for high-throughput matrix multiplication, achieving up to **32 parallel MAC operations per cycle** and optimized BRAM/DSP usage on the PYNQ FPGA.
- Verified and deployed on hardware using Vivado and Python testbench, demonstrating 100% functional accuracy across varying matrix sizes (4×4 to 32×32) and **reducing execution time by >10×** compared to CPU implementation.

Integer-Based Softmax and Layer Normalization for Quantized Neural Networks

Tech Stack: SystemVerilog, Fixed-Point Arithmetic, Vivado, PYNQ FPGA, Python

- Designed and verified integer-only Softmax and LayerNorm modules in SystemVerilog using pipelined fixed-point arithmetic, achieving full throughput at 100 MHz with **< 10 ns critical path**.
- Met LUT/FF/DSP utilization within **±30% of golden reference**, passed timing closure, and demonstrated 100% functional accuracy against Python golden models for multiple input configurations.

Hardware Resource Predictive Modeling using Classical ML, Convex Optimization, and Neural Networks

Tech Stack: Python, Scikit-learn, CVXPY, PyTorch, Matplotlib/Seaborn

- Developed predictive models to estimate FPGA hardware metrics (execution time, LUT, FF, RAM, DSP utilization) from systolic array parameters using Scikit-learn regression methods, CVXPY-based convex optimization, and PyTorch neural networks, achieving **RMSE reductions of up to 70%** with feature engineering.
- Implemented data preprocessing, z-score normalization, and engineered interaction features to improve model accuracy; optimized training pipelines with hyperparameter tuning and visualized performance using prediction-error and loss-curve plots.

RISC-V 5-Stage Pipelined Processor with UVM Verification

Tech Stack: SystemVerilog, UVM, Verilator, Vivado, Python, RISC-V ISA, PYNQ FPGA

- Designed and implemented a complete **RV32I 5-stage pipelined processor** (IF-ID-EX-MEM-WB) with full decode logic, ALU/branch unit, hazard detection, bypassing, and behavioral instruction memory; deployed on PYNQ-Z1 FPGA with correct BRAM mapping and timing closure.
- Built a **SystemVerilog UVM environment** (UVM sequences, drivers, monitors, scoreboards, functional coverage) to generate constrained-random RISC-V instruction streams, achieving **100% functional correctness** against golden traces and verifying pipeline/forwarding behavior under corner-case scenarios.

PROFESSIONAL EXPERIENCE

Silicon Design Engineer 2 | *Advanced Micro Devices, Inc. (AMD)* | *Markham, Canada*

Apr 2026 - Present

- Verified **Cache & Memory Management (CMM) blocks** within GPU/SoC architecture, ensuring functional correctness across complex memory and cache interactions.
- Developed and debugged **SystemVerilog/UVM-based testbenches**, creating reusable verification components and test scenarios for block-level validation.
- Defined and executed **verification plans**, including functional, performance, and power aspects, driving **verification closure within project timelines**.
- Implemented **constrained-random testing and functional coverage models** to improve coverage metrics and identify corner-case scenarios.
- Collaborated with RTL designers and architects to debug design issues, analyze failures, and ensure **high-quality design sign-off**.

Graduate Engineer Trainee | *Visteon India* | *Chennai, India*

Feb 2024 - Dec 2024

- Optimized SMT production processes through stencil design, reflow profile validation, and pick-and-place operations, **improving First-Time Through (FTT) and reducing scrap rates**.
- Conducted root cause analysis using **DFMEA** and **Six Sigma** methodologies, implementing corrective actions to reduce recurring defects and improve process stability.
- Prepared and maintained critical process documentation, including **PFMEA, Control Plans, SOPs, and Process Flow Diagrams**, ensuring compliance with automotive quality standards.
- Supported new product launches through **APQP** methodologies and **capability studies**, enhancing process readiness and launch execution.
- Utilized automotive testing and validation tools such as **CAN Protocol, Vector CANoe, and CANalyzer** for system validation, diagnostics, and troubleshooting.

Intern Trainee | *Visteon India* | *Chennai, India*

Jul 2023- Feb 2024

- Analyzed manufacturing workflows and supported process improvement initiatives to enhance **production yield** and reduce defects.
- Conducted **time-motion studies** on **SMT assembly lines** to identify bottlenecks and improve overall line efficiency.
- Assisted in **PCB assembly defect investigations** using the **8D problem-solving methodology**, gaining hands-on experience in quality control and root cause analysis.
- Created **work instructions, SOPs, and process documentation** in compliance with **IATF 16949** and **ISO 14001** standards.
- Collaborated with clients including **Tata, Mahindra, and Hyundai**, ensuring alignment between engineering and manufacturing requirements during production and validation activities.